

A Multimodal Time-of-Flight Based Haptic Proximity Feedback System with Directional Thermal and Vibrotactile Cues

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Abstract—This work presents a multi modal haptic proximity feedback system that utilizes a Time-of-Flight (ToF) distance sensor to detect nearby objects and convey feedback to the user using multi modal cues. The system is designed as a standalone dual-module architecture, consisting of a sensing unit integrated into a wearable spectacle frame and an actuation unit worn on the arm. The system incorporates a directional thermal feedback using three heating elements positioned at the left, center, right of the arm unit to indicate the relative direction of the detected obstacles. The embedded temperature sensing ensures safe operation within a controlled range of 15°C to 42°C. Additionally, a Voice Coil Actuator (VCA) is integrated to act as an alert vibrational cue when objects are within a critical proximity threshold for rapid user response. The proposed system enables intuitive perception of nearby objects without requiring visual attention or expensive camera-based systems. In addition to assistive navigation, this platform also has potential applications in human-machine wearable interfaces and spatial awareness systems.

I. INTRODUCTION

The ability to sense and respond to nearby obstacles is necessary to ensure safe navigation and interaction with the surroundings. For individuals with visual impairment, limited access to spatial information reduces mobility, independence, and overall quality of life. Although traditional and current assistive tools, such as white canes and guide dogs, provide critical support, they are limited in terms of range of detection, directional resolution, and the richness of feedback and cues. Recent advancements in wearable sensing and haptic technologies offer new opportunities to augment perception by translating environmental data into meaningful tactile-based signals [1]. Conventional approaches often rely on single-modality feedback, such as vibration, which can become difficult to interpret when mapping multiple dimensions such as direction and proximity.

Additionally, a range of assistive technologies have been proposed to address such challenges. Ultrasonic sensor-based systems are widely used for obstacle detection due to their integration simplicity and low cost, providing vibrotactile or auditory feedback to indicate proximity [2]. Camera-based systems and computer vision techniques offer richer environmental understanding but require higher computational resources and usually introduce latency [3]. More recently, Time-of-Flight (ToF) sensors have gained attention

for their compact form factor, accuracy, and suitability for real-time distance measurement in wearable applications [4]. In parallel, haptic feedback systems have largely focused on single-modality cues, most commonly vibration, to communicate obstacle presence or distance [5]. Multi modal haptic feedback systems address this limitation by distributing information across multiple sensory channels, thereby improving specificity and reducing cognitive load.

In this work, we propose a standalone multimodal haptic feedback system consisting of two integrated modules. The sensing module, implemented on a spectacle platform, combines a ToF sensor with a SEED XIAO module to continuously capture and communicate distance information. The actuation module, worn on the arm, receives this data and delivers feedback through a combination of directional thermal cues and vibrotactile alerts. Thermal feedback is spatially distributed to inform object direction, while vibration is used to signal critical proximity conditions requiring immediate attention. To ensure safe interaction, the system incorporates embedded temperature sensing and control-loop, maintaining actuator temperatures within a user-safe range.

II. BACKGROUND

A. Sensing System

Assistive technologies for visually impaired individuals increasingly rely on sensor-driven perception systems that map environmental information into interpretable sensory modalities. This method, often referred to as sensory substitution, enables users to perceive spatial information through tactile or auditory feedback. Among sensing technologies, Time-of-Flight (ToF) sensors provide accurate, low-latency distance measurements. The selected sensor provides accurate distance measurements within the range of approximately 20 to 1200 mm, with millimeter-level resolution. It operates over an I²C communication interface for seamless integration with the micro controller. The sensor supports fast sampling rates suitable for real-time applications while maintaining low power consumption, making it suitable for wearable systems. Moreover, its small form factor allows easy integration into the spectacle-mounted sensing module without compromising user comfort.

B. Haptic System

Haptic feedback is a widely explored modality in assistive systems due to its direct interaction with the user. Vibrotactile feedback is most commonly used to convey information through variations in frequency or intensity. However, relying

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on a single modality has historically limited the system's ability to communicate multiple dimensions of information simultaneously. Thermal feedback, although less explored, offers a complementary modality that can convey spatial cues through localized variations in temperature. Selected heating elements operate at 5V and require current in the range of 0.5–2 A, depending on the desired heating intensity. Temperature output is regulated using Pulse Width Modulation (PWM) to ensure smooth and gradual thermal variation. VCA supports a wide frequency range, between 50 Hz and 250 Hz, allowing modulation of vibration intensity through PWM or time-varying drive signals. VCA was chosen over conventional eccentric rotating mass (ERM) motors as it provides faster response time, better controllability, and reduced latency. The compact form factor and low power consumption make the VCA well-suited for integration into our wearable device. Combining multiple haptic modalities enables more instinctive perception by distributing information across distinct sensory channels.

C. Communication

Recent advances in wearable electronics and wireless communication, particularly Bluetooth Low Energy (BLE), allow for modular system architectures where sensing and actuation can be physically separated while maintaining real-time data transmission. This enables flexible and ergonomic designs for assistive devices.

TABLE I
SYSTEM COMPONENTS

Component	Model / Type	Function
ToF Sensor	VL53L0X	Distance measurement
Microcontroller (Tx)	XIAO nRF52840	Data acquisition and BLE transmission
Microcontroller (Rx)	Arduino Nano 33 BLE	Signal processing and actuator control
Thermal Actuators	Heating elements	Directional thermal feedback
Motor Driver	4-Channel MOSFET Driver	Switching control for heating elements
Vibrotactile Actuator	Voice Coil Actuator (VCA)	Critical proximity alert
Temperature Sensor	DHT11	Thermal safety regulation

TABLE II
SYSTEM PERFORMANCE METRICS

Metric	Value
Sensing Range	20–120 cm
Response Time	250 ms
BLE Latency	< 50 ms
Operating Temperature Range	36–42 °C
Directional Resolution	3 zones (Left, Center, Right)

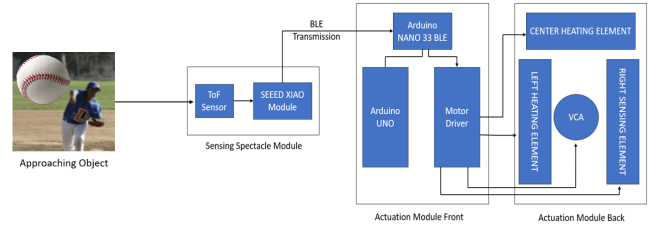


Fig. 1. System architecture of the multimodal haptic feedback system.

III. SYSTEM DESIGN

The proposed system is designed as a wireless dual-module architecture, consisting of a sensing spectacle module and an actuation arm-brace module.

The sensing module incorporates a ToF distance sensor along with a BLE-enabled SEED XIAO microcontroller. This module continuously measures the distance to nearby objects in the user's forward field of view and transmits the data wirelessly. The actuation module is implemented as an arm-mounted wearable device and contains a receiving Arduino NANO 33 BLE microcontroller, three thermal actuators positioned to represent left, center, and right directions, and a vibrotactile actuator for proximity alerts[Figure 1].

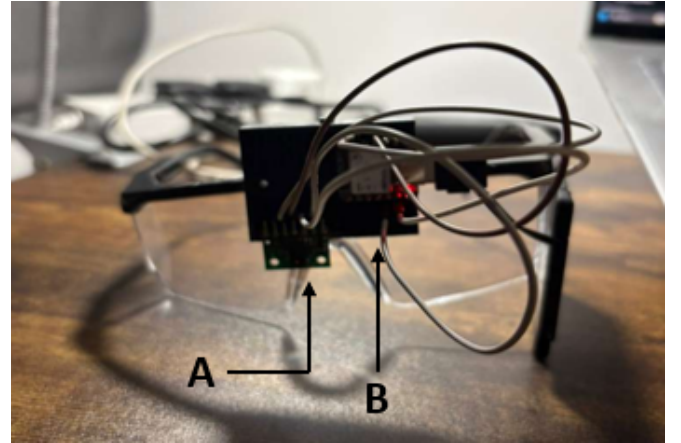


Fig. 2. Sensing module mounted on spectacles. (A) Time-of-Flight (ToF) sensor for distance measurement and (B) BLE-enabled microcontroller for data transmission.

The system is designed to map spatial information into haptic feedback such that directional cues are conveyed through localized heating, while urgent proximity conditions trigger vibration. The heating elements are driven by a MOSFET Trigger Switch 4 Channel Driver Module. Each channel of the driver is capable of handling currents up to 10A within an operating voltage range of 3.7–27V, allowing controlled switching of the heating elements. The MOSFET-based design is suitable for real-time thermal modulation as it ensures low switching losses, fast response times, and high efficiency.

The Time-of-Flight sensor measures the distance to nearby objects, while a microcontroller processes this data and

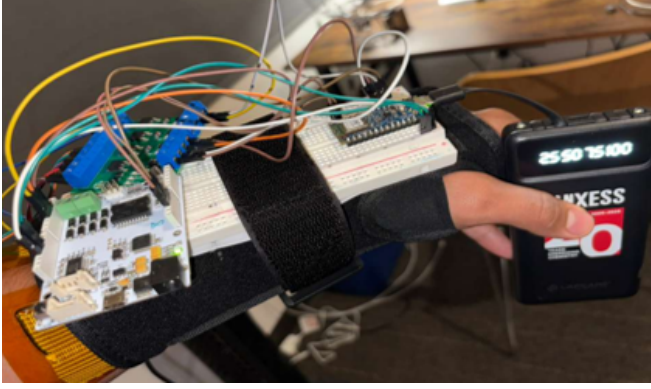


Fig. 3. Actuation module (outer view) showing embedded electronics, including the microcontroller, MOSFET driver, and power supply used to control the thermal and vibrotactile actuators.

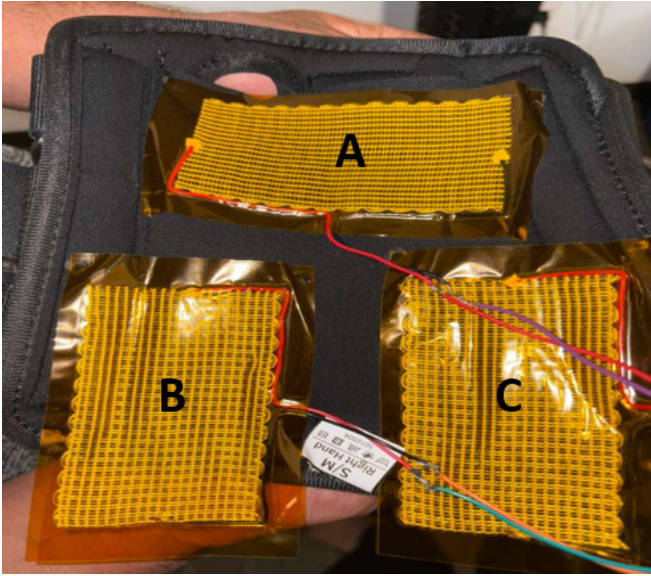


Fig. 4. Actuation module (inner view) showing placement of thermal elements. (A) Central thermal actuator, (B) left thermal actuator, and (C) right thermal actuator used for directional feedback.

transmits it wirelessly using Bluetooth Low Energy. On the receiving end, another microcontroller interprets the incoming data and uses given parameters to drive the haptic outputs. Thermal actuators are used to indicate direction, while a vibration motor provides alerts for close proximity threshold. Temperature sensing is integrated to ensure that the thermal feedback remains within a safe range, maintaining the operating range under 42°C .

IV. METHODS

The system maps distance measurements obtained from the ToF sensor into distinct haptic outputs accounting for both direction and proximity. The measured distance is categorized into discrete zones corresponding to different levels of interaction. Objects beyond a predefined threshold produce no feedback to avoid unnecessary stimulation. As objects enter the active sensing range, directional thermal

feedback is generated using three spatially distributed actuators corresponding to left, center, and right arm positions.

A. Feedback Mapping Strategy

Directional information is determined based on the relative orientation of the sensing module, and the corresponding appropriate heating element is activated. The intensity of thermal feedback is modulated using Pulse Width Modulation (PWM), with closer objects resulting in higher duty cycles and increased thermal intensity. This creates a continuous gradient of feedback that allows users to perceive changes in distance without abrupt transitions.

When the detected distance falls below a critical threshold, a vibrotactile alert is triggered using the VCA. This serves as an alert signal indicating immediate proximity and potential risk. By separating directional and urgency information across thermal and vibrotactile modalities, the system reduces ambiguity and improves interpretability.

Let d_{far} denote the upper threshold beyond which no feedback is provided, d_2 denote an intermediate threshold separating low and medium thermal feedback, and d_{crit} denote the critical proximity threshold at which vibrotactile alerts are triggered. For our work, these thresholds are empirically selected as $d_{\text{far}} = 100$ cm, $d_2 = 50$ cm, and $d_{\text{crit}} = 20$ cm. For $d > d_{\text{far}}$, no feedback is generated to avoid unnecessary stimulation. When $d_2 < d \leq d_{\text{far}}$, low-intensity thermal feedback is applied within the range of $36\text{--}39^{\circ}\text{C}$. For $d_{\text{crit}} < d \leq d_2$, medium-intensity thermal feedback is generated within $39\text{--}42^{\circ}\text{C}$. Finally, for $d \leq d_{\text{crit}}$, a vibrotactile alert is activated to indicate immediate proximity.

TABLE III
FEEDBACK MAPPING STRATEGY

Distance Range	Feedback Type	Intensity
$d > d_{\text{far}}$	No feedback	None
$d_2 < d \leq d_{\text{far}}$	Directional thermal (L/C/R)	Low ($36\text{--}39^{\circ}\text{C}$)
$d_{\text{crit}} < d \leq d_2$	Directional thermal (L/C/R)	Medium ($39\text{--}42^{\circ}\text{C}$)
$d \leq d_{\text{crit}}$	Vibrotactile (VCA) alert	High (frequency-modulated vibration)

B. Thermal Control and Safety

Thermal feedback is regulated through a closed-loop control system to ensure safe and reliable operation of the heating elements. Each thermal actuator is integrated with a temperature sensor that continuously measures the actuator surface temperature, denoted as T . The system enforces safety constraints such that the temperature remains within a predefined safe range:

$$T_{\min} \leq T \leq T_{\max}$$

where $T_{\min} = 36^{\circ}\text{C}$ and $T_{\max} = 42^{\circ}\text{C}$.

The heating elements are driven using Pulse Width Modulation (PWM), where the duty cycle α determines the effective thermal output. The duty cycle is initially determined

based on the distance-to-feedback mapping. The actuator is automatically disabled when the temperature exceeds the safe threshold.

To improve user comfort and avoid abrupt thermal transitions, a gradual ramping function is applied to the duty cycle:

$$\alpha(t) = \alpha_{\text{safe}} (1 - e^{-kt})$$

where k is a constant that controls the rate of temperature increase. This results in smooth and controlled heating behavior.

V. RESULTS

The proposed multimodal haptic feedback system was evaluated through a combination of system-level testing and a small-scale pilot study. The results focus on assessing the reliability of the sensing module, interpretation of feedback, and user response.

A. System Performance

The ToF sensor demonstrated accurate distance measurements within the operating range of 20–120 cm. The system maintained consistent performance with minimal measurement noise under controlled indoor conditions. BLE communication enabled real-time data transmission between the sensing and actuation modules, with observed latency below 50 ms.

The thermal actuators successfully produced distinguishable temperature ranges corresponding to low and medium intensity feedback as explained above. The use of PWM allowed smooth modulation of thermal output, and the implemented safety constraints ensured that actuator temperatures remained within the predefined safe limits. The vibrotactile actuator provided immediate and noticeable alerts under critical proximity conditions.

B. Pilot User Study

A preliminary user study was conducted with five participants to evaluate the usability, interpretability, and ergonomics of the multimodal feedback system. Participants were blindfolded and asked to identify the direction (left, center, right) as well as proximity level of objects based solely on haptic feedback.

TABLE IV
PILOT STUDY USER RATINGS (N = 5)

Metric	Mean Rating	Std. Dev.
Usability	4.2	0.4
Feedback Clarity	4.4	0.5
Ergonomics	3.8	0.6

The results of the pilot study indicate that participants were able to easily interpret both directional and proximity cues using the proposed multimodal feedback system. The system achieved high user ratings in feedback clarity (mean = 4.4/5), reflecting the ability of participants to clearly distinguish between directional thermal signals and vibrotactile alerts.

Usability was also rated highly (mean = 4.2/5), suggesting that users were able to quickly understand and respond to the feedback with minimal training. Ergonomics received a slightly lower rating (mean = 3.8/5), indicating that while the system was generally comfortable, there is scope for improvement in form factor.

Participants demonstrated strong performance in identifying directional cues and proximity, supported by the high feedback clarity scores. The inclusion of vibrotactile alerts contributed to improved recognition of critical proximity events. The average response time for interpreting and reacting to feedback was approximately 1.5 seconds, indicating that the system enables near real-time interaction.

The combination of thermal and vibrotactile cues allows users to reliably perceive spatial direction and urgency and simultaneously maintaining a low cognitive burden.

VI. DISCUSSION

The results obtained during the pilot study demonstrate the feasibility and effectiveness of the proposed multimodal haptic feedback system in conveying both spatial and proximity information. The combination of thermal and vibrotactile feedback enabled users to interpret directional cues and urgency with relatively high accuracy. However, several areas for further development and validation still remain.

One key area for improvement lies in the ergonomic design of the sensing and actuation module. While the system was able to achieve high usability and feedback clarity scores, the slightly lower ergonomics rating suggests that refinements in form factor and material selection could enhance long-term user comfort. Future iterations could explore flexible electronics in the form of a fPCB and more flexible heating elements, soft materials, and more compact actuator integration to improve the overall user experience.

The pilot study was limited to a small sample size of only five participants under controlled indoor conditions. To more rigorously evaluate the system, a larger-scale study is required. A proposed study would involve 20–40 participants, including individuals with visual impairments, navigating structured indoor environments with obstacles placed at varying distances and orientations. The system’s performance can be evaluated across different feedback modalities, response time to feedback and comfort.

Key evaluation metrics for the larger study would include navigation accuracy, obstacle avoidance success rate, response time, and cognitive load, which can be assessed using a standardized questionnaire. Additionally, user preference and long-term usability can be evaluated through extended trials.

Overall, the proposed system provides a promising foundation for assistive haptic technologies. With further refinement in hardware design and comprehensive user validation, it has the potential to enhance spatial awareness and mobility for individuals with visual impairments.

VII. CONCLUSIONS

This work presented a multimodal haptic feedback system that combined directional thermal cues with vibrotactile

TABLE V
PROPOSED LARGE-SCALE STUDY DESIGN

Parameter	Description
Participants	20–40 individuals (visually impaired users)
Environment	Indoor navigation course with controlled obstacles
Study Conditions	Thermal-only, Vibrotactile-only, Multimodal
Tasks	Obstacle detection, direction identification, navigation
Metrics	Accuracy, response time, collision rate, path efficiency
Cognitive Load	Questionnaire
User Feedback	Usability, feedback clarity, ergonomics
Outcome Measure	Comparison of performance across feedback modalities

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alerts to enhance spatial awareness. The results demonstrate that distributing information across multiple sensory channels improves interpretability and reduces cognitive effort compared to single-modality approaches.

The proposed system has potential applications in assistive technologies for visually impaired individuals, wearable navigation aids, and human–machine interaction systems where intuitive feedback is critical. The integration of compact sensing, wireless communication, and multimodal actuation highlights the feasibility of developing lightweight and scalable wearable solutions.

Future work will focus on improving the ergonomic design, optimizing feedback mapping strategies, and validating the system through larger-scale user studies. With further refinement, the proposed approach can contribute to more effective and user-friendly assistive devices for real-world deployment.

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